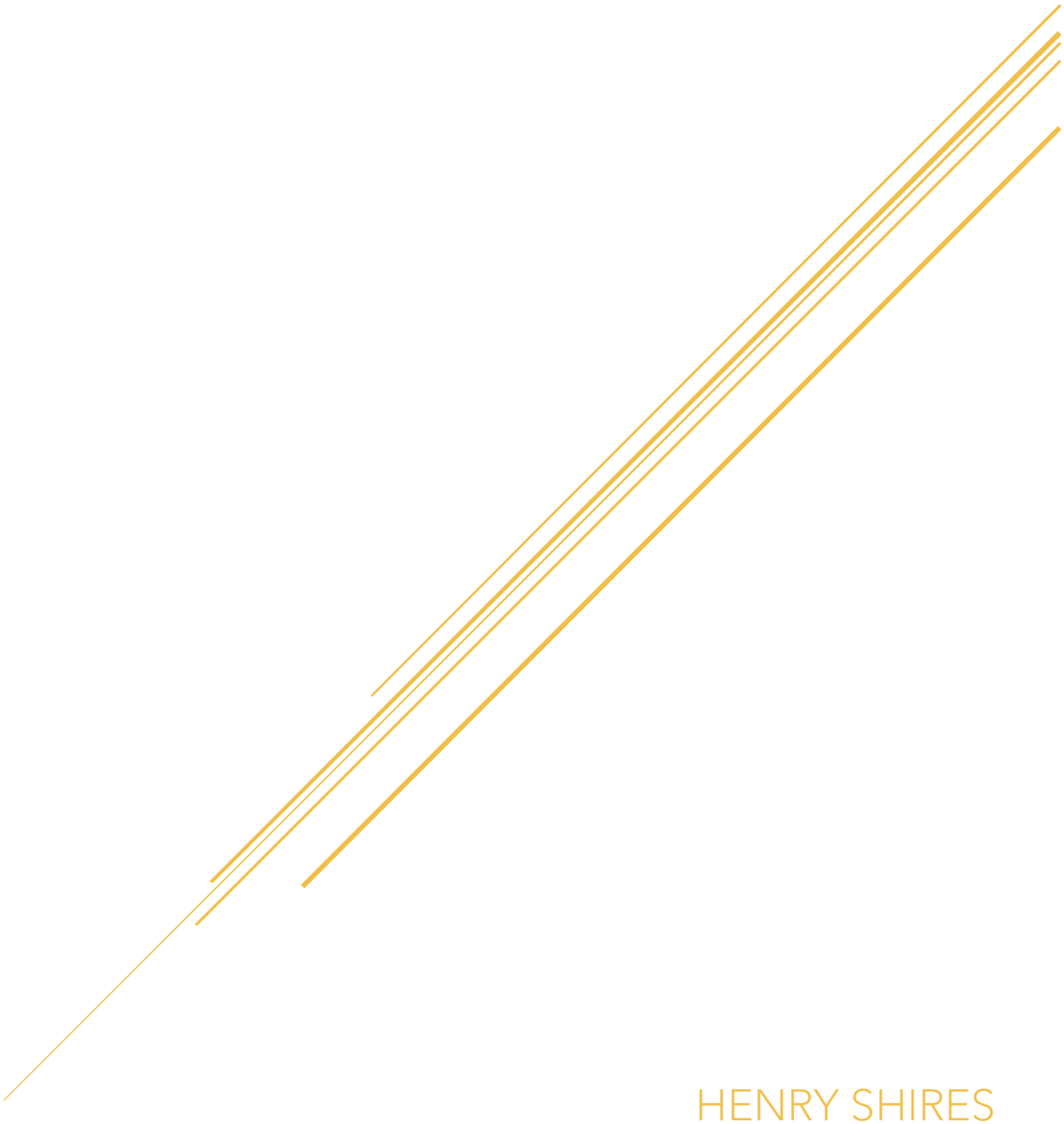
AN OVERVIEW OF COMMUNICATING WITH CYSAT-1

SPRING 2023 COMMUNICATION HARDWARE REPORT



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Disclaimer

This report was written based on the hardware for CySAT-1 as of March 2023. Since then, the team has adapted an alterative approach to implementing satellite communications on the ground via a Software-Defined Radio (SDR) and GNU Radio modules due to the digital packet transmission and reception limitations of physical ham radio devices. However, many of the principles and the hardware devices onboard the satellite covered in this document have not changed. For inquiries on the updates made to CySAT-1 in April through August of 2023, contact the author, Henry Shires, at hcsgeek@gmail.com.

Introduction

Nearly 10 years ago, Iowa State University engineering students conceived and designed a *NASA CubeSat* capable of being launched on a rocket into space. The National Aeronautics and Space Administration (NASA) defines a *CubeSat* as a type of rectangular-prism satellite with specific dimension constraints created to simplify the process of short-term satellite engineering (Chin 1). The CubeSat standard is part of NASA's *CubeSat Launch Initiative (CSLI)*, an educational program aimed at enabling institutions and student engineers to quickly design, build, test, and launch their satellites into space by providing a free ride on a rocket in exchange for a copy of the satellite's mission results (Chin 2). These constraints create a standard to unify the design of each satellite NASA receives via the CSLI, enabling efficient handoff and launch. It also provides an easy way for engineers to buy hardware and tools made explicitly for



Figure 1: Examples of a 1U CubeSat (left) and 3U CubeSat (right) (Chin 4)

the CubeSat standard. Figure 1 on page 1 illustrates two examples of CubeSats.

The Make to Innovate program, a project-based class through the Department of Aerospace Engineering, sponsors a CubeSat program called *CySat* (short for *Cyclone Satellite)*, designing and launching a CubeSat to measure soil moisture contents of the Earth from Low-Earth Orbit. According to The European Space Agency, *Low-Earth Orbit* is the status of moving around the Earth at a height between 160 and 1000 kilometers above the Earth's surface ("Low Earth orbit"). CySat's scientific mission is onboard *CySat-1*: a *3U* or 10 cm x 10 cm x 34 cm satellite with an anticipated launch date of Summer 2024 (see Figure 2). Each "U" represents one 10 cm x 10 cm x 11 cm cube stacked on each other to form a rectangular prism shape (Chin 4). If accepted upon handoff to NASA, CySat-1 will be Iowa State University's first-ever successfully launched CubeSat into space.

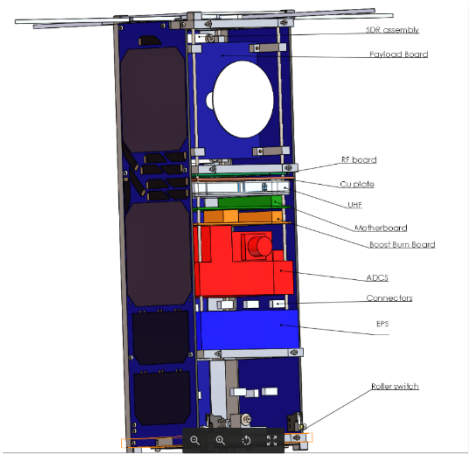


Figure 2: A digital diagram of the 3U CySat-1 CubeSat and its subsystems (original work)

To carry out its mission, CySat-1 requires several internal electronic and mechanical subsystems and a communications interface to wirelessly link its onboard systems to the ground, enabling the satellite to receive operational commands and transmit valuable measurement data while in space. Satellite teams typically perform wireless communication with the use of *amateur radio* (Tanaka 887). Amateur radio is a popular mode of communicating between electronics using radio waves within a unique frequency range permitted by the Federal Communications Commission (FCC) ("What is Ham Radio"). Radio is a widely adopted technology involving transmitting and receiving types of electromagnetic waves that carry valuable information. According to the National Association for Amateur Radio (ARRL), amateur radio, more commonly known as *ham radio*, is used by almost anyone, from hobbyists to truck drivers and other professionals, to talk to other radio users for fun or to communicate with electronic devices ("What is Ham Radio"). CubeSats with onboard amateur radio-compatible transmitters can utilize this sound

technology, as described in the "Component Summary" section on page 3. And many CubeSat projects adopt amateur radio as their methods of communication, such as Takushi Tanaka and Yoshiyuki Kawamura's *FITSAT-1* (Tanaka 887).

This document will describe the mechanism and components required to successfully test a wireless amateur radio link between CySat-1 and a *ground station* (see the "Ground Station Radio" section on page 5). CySat-1 utilizes numerous devices from *EnduroSat*, a company specializing in manufacturing and distributing parts for high-end CubeSats and other software and equipment ("EnduroSat"). While the satellite requires many devices to function correctly, this document will only focus on the necessary components for the satellite to communicate to a ground station and vice versa.

Component Summary

The following components contribute to sending and receiving commands between the ground and the satellite. Each part is compatible with the other and the CubeSat standard, making it easy and efficient to design an operational satellite with them.

Satellite Onboard Computer (OBC)

The satellite requires a centralized computer to act as the "brains" of the mission. Like a company's general manager directs its employees, the satellite needs a central unit to process information and direct tasks to other subsystems, such as toggling signals and triggering important mission events. These tasks are the responsibilities of the Onboard Computer (OBC), a specialized microcomputer that connects with the other electronic subsystems in the satellite and runs the main mission application. All subsystems onboard CySat-1 connect to the OBC and receive instructions from it on what to do and when. Additionally, commands from the ground are sent to the OBC to be

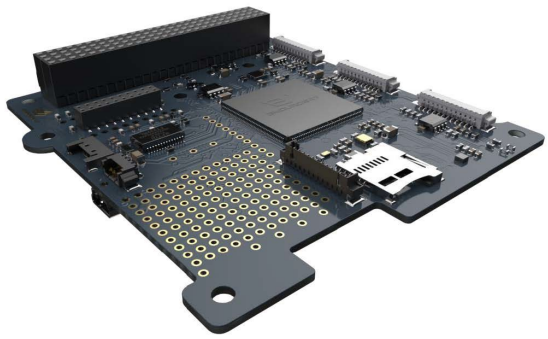


Figure 3: A top-down view of the EnduroSat OBC Type II ("CubeSat Onboard")

processed and executed ("CubeSat Onboard"). Figure 3 illustrates the physical OBC component, including the various computer chips and pins in its design.

Satellite Ultra-high Frequency (UHF) Antenna

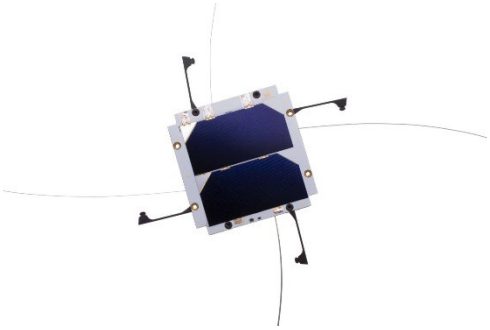


Figure 4:A view of the UHF Antenna II ("CubeSat UHF Antenna") from the solar panel side and in the deployed state.

An antenna device is required to generate and pick up radio signals to communicate over radio waves. An antenna uses the properties of copper wire to transmit and receive these types of electromagnetic signals reliably. According to EnduroSat, the *UHF Antenna II* used on CySat-1 is a circular copper wire antenna that fits within the satellite and operates between radio frequencies 435-438 Megahertz (MHz) or millions of cycles per second ("CubeSat UHF Antenna"). A frequency is the speed at which radio waves pulse or oscillate. The higher the frequency, the faster the waves oscillate, enabling penetration of large objects or creating stronger signals over large distances. This range of frequencies is considered *ultra-high frequency* (UHF), as the defined scope of UHFs is 300 MHz – 300 gigahertz (GHz) or 300 billion cycles per second. Radio communication works by having a transmitter on one end and a receiver on the other end, with both ends using antennas or specially designed copper wire to pick up the transmitted signal over a specific frequency. Since the Federal Communications Commission (FCC) regulates the frequency a user can operate on, EnduroSat provides a range of possible frequencies to configure ("CubeSat UHF Antenna"). CySat's licensed frequency by the FCC is *436.375 MHz*.

Satellite UHF Transceiver



Figure 5: UHF Transceiver Type II ("CubeSat UHF Transceiver")

The antenna must connect to a computer called a *transceiver* to control waves transmitted and received. The transceiver is responsible for converting radio waves into understandable digital information and any

information into radio waves it wants to send ("CubeSat UHF Transceiver"). The satellite will talk to all potential transceivers over the radio, so the protocol to convert data into radio waves must be consistent. The transceiver uses two radio standards: AX.25 and GFSK. *AX.25* is a designated protocol for sending digital signals from one radio to another (Beech 1) used by amateur radio operators. *Gaussian Frequency Shift Keying,* or GFSK, is an algorithm managed by the AX.25 protocol to *encode* or translate digital bits of information into an analog radio wave in the transmitting radio system ("Definition of GFSK"). The combinations of these two standards enable the receiving radio system to properly *decode* or translate the sent analog waves back into digital bits, the form factor used by the computers on either end of the transmission. This process is comparable to having a conversation in the same language, as both participants know how to speak and understand the language allowing for proper communication. The *EnduroSat UHF Transceiver Type II*, as depicted in Figure 5, can communicate with both AX.25 and GFSK protocols and is compatible with the UHF Antenna Type II in Figure 4, completing the satellite end of the communication link ("CubeSat UHF Transceiver").

Ground Station Amateur Radio



Figure 6: Kenwood TMD700A Radio and TNC ("KENWOOD")

After the satellite is in space, there is no physical way to interact with it and ensure it's functioning correctly except via radio communication. On the ground, CySat has designed a *ground station* or a radio communication interface that includes a radio and personal computer (PC) with a specialized program to read the transmissions from the ground station radio easily. For CySat-1, a Kenwood TMD700A radio unit, as pictured in Figure 6, is used to receive any messages from the satellite and convert them into readable digital characters, as it provides an easy setup to test the satellite's communication hardware before being in space. According to *Kenwood*, the manufacturer of the TMD700A, the radio features a *Terminal Node Controller* (TNC), which acts as the ground station's version of the UHF transceiver on the satellite ("Kenwood Owner" 1). The TNC is responsible for sending and receiving packets from the PC to the

satellite over the radio's antenna via encoding with AX.25 and GFSK. Additionally, the TNC is responsible for communicating with the ground station PC, as explained in the "Computer Software" section below on page 6.

Computer Software

The ground station now needs a way to send and receive signals at the user's request on the ground station end of the communication. For CySat, two options are available:

1. The team can utilize a custom command terminal called *PuTTY*, designed to send commands to and receive messages from external devices ("PuTTY").
2. The team can run the *CloneComm* user interface: a custom CySat ground station program designed to simplify sending commands and receiving messages.

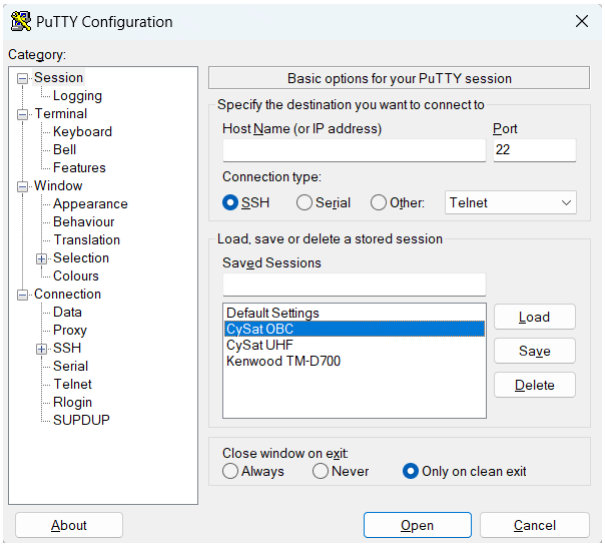


Figure 7: A screenshot of the PuTTY configuration window ("PuTTY")

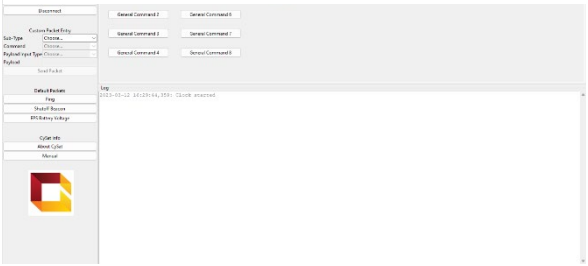


Figure 8: A screenshot of the CloneComm user interface designed by the CySat team (original work)

Regardless of the program chosen, both programs rely on a USB cable connection to the Kenwood radio. USB relies on a communication standard called *Serial,* which breaks pieces of data into *packets* easily deliverable to various computer devices (Peña). Packets are segments of one large amount of digital information formatted to be quickly sent via a communication protocol (examples in this document are Serial and AX.25). Much like mail delivery to a respective address, Serial breaks large chunks of data into pieces and delivers them safely to the destination. A Universal Asynchronous Receiver Transmitter (UART) device is built into the

PC and Kenwood TNC to facilitate Serial communication over USB. This communication link is comparable to the radio communication link between the entire station and the satellite but at a much smaller scale than radio (Peňa).

Process

Once all ground station and satellite components are correctly configured and enabled, communication begins in a particular order. This process relates to one long invisible wire connecting the ground to the satellite in space and back, where the invisible wire is the use of radio waves. First, the user types a command into the interface of choice (PuTTY or CloneComm) formatted in ordinary English. This command will then be sent along the flow chart, as seen in Figure 9. The command gets converted into Serial packets to get sent along USB to the ground station radio. The PC UART module will format and send the Serial packets to the radio.

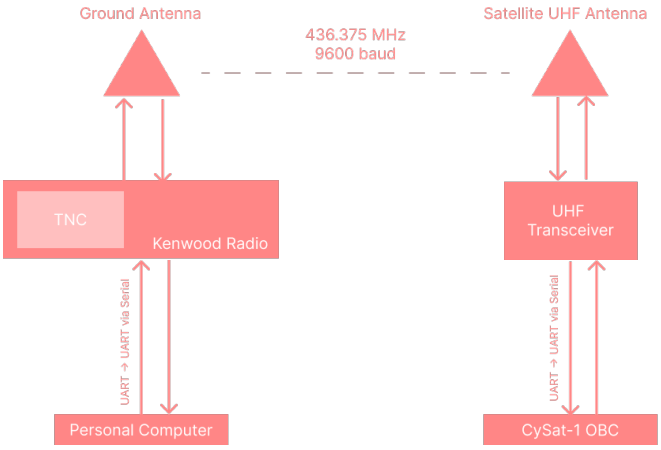


Figure 9: A flow chart of the CySat-1 communication process (original work)

Next, the radio's receiving UART will disassemble the packets to send over the antenna. The radio will pick up the transmission to send and give the packets to the TNC to encode into the AX.25 protocol using GFSK. The TNC will pass the encoded packets to the antenna to transmit on the assigned frequency. After some time, the transceiver on the satellite will pick up the incoming packets over the same frequency and begin decoding them. These get converted from GFSK analog signals into digital packets for the Onboard Computer (OBC) to interpret as commands or actions.

Finally, if programmed to do so, the OBC will command the UHF transceiver on the satellite to send a proper response. Depending on the received command from the ground, this could include a simple confirmation that the satellite received the packet or a return set of packets with information from the satellite's instruments (measurements, other data, etc.). This data will be encoded by the transceiver using AX.25 and GFSK, sent to the antenna, transmitted over the designated frequency, then picked up on the ground assuming the satellite is directly above the ground station. The ground station radio will receive the transmission and decode the signals into digital packets via the TNC. The TNC will then transfer the digitized packets to the PC to convert them back into readable characters the users on the ground can observe. This successful *roundtrip* or communication from the ground station to the satellite and back to the ground station completes the process.

Conclusion

By utilizing the pre-existing amateur radio technology, CubeSats are relatively straightforward to communicate with and can enable practical tools for controlling devices in space from the ground. During CySat-1's three-month mission in space, theoretically, any student at Iowa State interested in saying "hello" to the satellite can do so in seconds utilizing the communication mechanism described, given the satellite is in the correct orbit position above campus. Radio waves enable communication across distances that seem unfathomable with incredible quality and reliability, ensuring CySat's mission remains possible. Utilizing NASA's CubeSat standard, CySat can design, build, and deploy a powerful satellite capable of precise measurements to confirm scientific hypotheses and provide real-world, valuable data to NASA upon mission completion. Through these developed technology standards, CySat can create a meaningful spacecraft the size of a water bottle and talk to it from an incredible 1000 kilometers below, a feat once thought impossible by students from only a few years ago.

Glossary

3U: A 10 cm x 10 cm x 34 cm CubeSat satellite.

AX.25: A designated protocol for sending packets from one radio to another used by amateur radio operators.

Amateur (Ham) Radio: A popular mode of communicating between electronics utilizing the concept of radio waves within a unique frequency range permitted by the Federal Communications Commission (FCC).

Antenna: A specially designed copper wire to send or pick up a radio signal transmitted over a specific frequency.

CloneComm: a custom CySat ground station program designed to simplify sending commands and receiving messages.

CubeSat: A type of rectangular-prism satellite with specific dimension constraints created to simplify the process of short-term satellite engineering.

CubeSat Launch Initiative (CSLI): An educational program that enables institutions and student engineers to quickly design, build, test, and launch their own satellites into space by providing a free ride on a rocket in exchange for a copy of the satellite's mission results.

Decode: To translate a received signal, usually analog radio waves, into a readable format, typically digital bits.

Encode: To translate data, typically digital, into an easy-to-send format, such as analog radio waves.

Ground Station: A radio communication interface that includes a radio and personal computer (PC) with a specialized program installed, enabling users on the ground to talk to and hear from the satellite once in space.

Gaussian Frequency Shift Keying (GFSK): An algorithm managed by the AX.25 protocol to encode or translate digital bits of information into an analog radio wave in the transmitting radio system.

GNU Radio: An open-source program designed to create signal processing programs that run on software-defined radios and in signal simulations.

Low-Earth Orbit: The status of moving around the Earth at a height between 160 and 1000 kilometers above the Earth's surface**.**

Onboard Computer (OBC): A specialized microcomputer that connects with the other electronic subsystems in the satellite and runs the main mission application.

Packet: Segments of one large piece of digital information formatted to send via a communication protocol.

PuTTY: A customizable computer terminal designed to send commands to and receive messages from external devices.

Radio: A concept involving the transmission and reception of particular types of electromagnetic waves that usually carry helpful information.

Roundtrip: A successful communication link from the ground station to the satellite and back to the ground station.

Serial: A data transmission protocol that breaks pieces of data into *packets* easily deliverable to various computer devices.

Software-Defined Radio (SDR): A computer or embedded system that implements typical analog components in a radio or other signal devices and can be controlled by software.

Terminal Node Controller (TNC): A device in the ground station radio responsible for sending and receiving packets from a PC to the satellite over the radio's antenna via encoding with AX.25 and GFSK.

Universal Asynchronous Receiver/Transmitter (UART): A device built into the ground station PC and Kenwood TNC that facilitates Serial communication over USB.

Ultra-High Frequency (UHF): A radio frequency within 300 MHz – 300 gigahertz (GHz) or 300 billion cycles per second.

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